Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Keira Fidgeon	kfidgeon@udel.edu
Friend (optional)	Sydney Whittaker	

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

An idea I had is to create a website that uses user's input on their favorite tropes and genres of romance books and pick a random book for the reader's next read for indecisive readers.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

romance.io, TropeQuest.com, https://bookbrood.com/?utm_source=chatgpt.com

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would want this web application to help indecisive readers choose their next book by recommending romance novels based on their favorite genres and tropes. Users would select the types of stories they enjoy, and the website would show them books that match their preferences, making it easier to discover their next read.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal of my program is to take user input about their favorite romance genres and tropes, using a Book dataclass to organize book information, and use loops and conditionals to filter a list of books for matching genres and tropes before randomly selecting a book to recommend.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to select their favorite romance genres and tropes, such as a contemporary romance and friends to lovers.

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3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will compare user's selections with each book's stored genres and tropes, filter out books that do not match, and organize the remaining results.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a book suggestion randomly based on their inputs of what they like.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The main things my program needs to represent are books, genres, tropes, and user preferences.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need are the book's title, author, genres, and tropes Details I will ignore are the book's page count, cover design, publisher, and physical size

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult part of this project will be figuring out how to filter books based on the user's selected genres and tropes and then randomly choose one of the matching books. I will also need to make sure the website handles situations where no books match the user's preferences.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will ensure I understand the generated code by asking the LLM to explain the code and break down anything I don't understand. I will also test the code to make sure it works correctly and compare it with what I have learned in class.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal for this project is to learn how to use drafter and make something fun out of it.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to create a website that randomly picks a romance book for indecisive readers based on their favorite genres and tropes. What are some existing websites that do something similar, and what are some simple features I could include in my website for an intro to computer science project?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

AI suggested looking at websites like Romance.io, Book Brood, and TheStoryGraph, which helps readers discover books based on their interests. My website could allow users to choose their preferred genres and tropes to find books that suit their interests. It would include a feature that randomly recommends a book based on their selections, along with a page showing information about the chosen book, including its title, author, a short description, and its main tropes. Users would also be able to click a button to generate a new recommendation whenever they want to discover something different.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think including a description for every book might be too complicated and time consuming for an introductory programming project because I would have to find and enter information for each book. Could we simplify this feature by displaying only the book's title, author, genre, and tropes instead.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project idea has become more focused because I decided to have the website randomly pick one romance book based on the user's favorite genres and tropes instead of giving multiple recommendations. I also simplified the book information to include only the title, author genre, and tropes instead of a full description.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use my peers and professors if I get stuck. If I get stuck using AI I will keep going or ask it questions when in need.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will go line by line myself and change whatever is causing the error.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to create a website that randomly picks a romance book for indecisive readers based on their favorite genres and tropes. What are some existing websites that do something similar, and what are some simple features I could include in my website for an intro to computer science project?

Your initial prompt looks good! The next step is to summarize the ideas from the brainstorming conversation for Question 2: The Response.

The AI suggested looking at websites like Romance.io, Book Brood, and The StoryGraph, which help readers discover books based on their interests. Some simple features I could include are letting users select their favorite genres and tropes, randomly picking one matching book, and displaying the book's title, author, description, and tropes. I could also add a button that lets users pick another book if they want a different one.

This keeps your response focused on the existing websites and features that are realistic for an introductory computer science project.

I think including a description for every book might be too complicated and time consuming for an introductory programming project because I would have to find and enter information for each book. Could we simplify this feature by displaying only the book's title, author, genre, and tropes instead.

That's a good simplification! You can use this as your response for Part 5, Question 3: The Pushback.

I think including a description for every book might be too complicated and time-consuming for an introductory programming project because I would have to find and enter information for each book. Could we simplify this feature by displaying only the book's title, author, genre, and tropes instead?

This keeps your project manageable while still giving users the information they need to decide whether they want to read the randomly selected book.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again.

Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.